Sabbir Ahmed

Electrical & Electronic Engineering Undergraduate

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SUMMARY

Electrical and Electronic Engineering undergraduate at United International University with hands-on experience in embedded systems, IoT, and hardware testing through competitive robotics (UIU Mars Rover Team) and independent microcontroller projects. Comfortable working across the full stack of a hardware build: circuit design, firmware, wireless communication, and technical documentation.

SKILLS

Languages & Tools: C,Python (basic), MATLAB & Simulink, GitHub,

Hardware & Embedded: Microcontrollers (ESP32, Arduino), Analog Circuit Design, IoT, Hardware Testing & Troubleshooting

Other: Research & Technical Documentation, Teamwork & Cross-functional Collaboration

EXPERIENCE

Electrical Sub-Team Member | UIU Mars Rover Team | *2024 – 2025*

- Designed, developed, tested, and integrated rover electrical systems as part of a multi-disciplinary engineering team building reliable rover subsystems.
- Performed hardware validation and troubleshooting for international rover competitions, contributing to research and technical problem-solving under competition deadlines.
- Helped the team win the University Rover Challenge (URC) 2025, earning Best Asian Team honors for the fourth consecutive year, and contributed to a strong result at the Anatolian Rover Challenge (ARC).

Researcher | CENTeR — Center for Emerging Networks and Technologies Research | *2025 – Present*

- [Add 1–2 lines here on your actual research focus/output — e.g., topic area, tools used, any papers or reports produced. Left as a placeholder since it wasn't detailed on your GitHub/LinkedIn.]

Batch Ambassador | Dept. of EEE, United International University | *2024 – 2026*

- Served as Batch Ambassador for EEE Day 2024, liaising between faculty and students to organize and execute the departmental event.
- Coordinated event activities and cross-team communication, contributing to successful program delivery for the department.

PROJECTS

ESP32 RC Car with Joystick Control | github.com/SabbirLab/Sabbir-Ahmed

- Built a wireless remote-controlled car using an analog joystick transmitter and ESP32 receiver communicating over ESP-NOW.
- Designed and implemented the transmitter and receiver firmware for real-time motor control based on joystick input.

2×3 Matrix Keypad Interface (ESP32) | github.com/SabbirLab/2x3-Keypad-Using-ESP32

- Interfaced a 2×3 matrix keypad with an ESP32 microcontroller using the Keypad library to detect and process keypresses in real time, outputting results to the serial monitor.

Series RLC Resonant Circuit Analysis (MATLAB) | github.com/SabbirLab/Sabbir

- Modeled a series RLC resonant circuit (C = 1F, L = 2.7 mH) driven by a 5 Vpp, 1 kHz sine wave, sweeping capacitance from 0.1F to 20F to analyze resonance behavior in MATLAB.

EDUCATION

Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering (EEE)

United International University | 2024 – Present

Diploma in Electrical Technology

Gopalganj Polytechnic Institute | 2018 – 2022 | CGPA: 3.77 / 4.00